

- 9 (a) e.g. reduced gain
increased stability
greater bandwidth or less distortion
(allow any two sensible suggestions, 1 each, max 2) B2 [2]
- (b) (i) V^- connected to midpoint between resistors
 V_{OUT} clear and input to V^+ clear B1
B1 [2]
- (ii) gain = $1 + R_F/R$
 $15 = 1 + 12000/R$ C1
 $R = 860 \Omega$ A1 [2]
- (c) graph: straight line from (0,0) to (0.6,9.0) B1
straight line from (0.6,9.0) to (1.0,9.0) B1 [2]
- (d) either relay can be used to switch a large current/voltage M1
output current of op-amp is a few mA/very small A1 [2]
or relay can be used as a remote switch (M1)
for inhospitable region/avoids using long heavy cables (A1)

- 10 (a) e.g. large bandwidth/carries more information